

Quantifying and Attributing Polarization to Annotator Groups

Anonymous ACL submission

Abstract

Current annotation agreement metrics are not well-suited for inter-group analysis, are sensitive to group size imbalances and restricted to single-annotation settings. These restrictions render them insufficient for many subjective tasks such as toxicity and hate-speech detection. For this reason, we introduce a quantifiable metric, paired with a statistical significance test, that attributes polarization to various annotator groups. Our metric enables direct comparisons between heavily imbalanced sociodemographic and ideological subgroups across different datasets and tasks, while also enabling analysis on multi-label settings. We apply this metric to three datasets on hate speech, and one on toxicity detection, discovering that: (1) Polarization is strongly and persistently attributed to annotator race, especially on the hate speech task. (2) Religious annotators do not fundamentally disagree with each other, but do with other annotators, a trend that is gradually diminished and then reversed with irreligious annotators. (3) Less educated annotators are more subjective, while educated ones tend to broadly agree more between themselves. Overall, our results reflect current findings around annotation patterns for various subgroups. Finally, we estimate the minimum number of annotators needed to obtain robust results, and provide an open-source Python library that implements our metric.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity, and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable (United Nations, 2025; Dioum, 2025). However, annotations for such

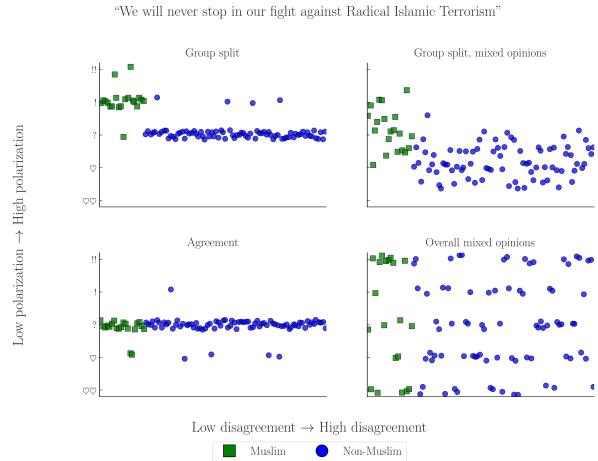


Figure 1: Simulated experiments for four scenarios on a hate speech detection task. Depending on the annotators, there may be a clear pattern where the split in annotations (polarization) can be attributed to a specific group (Muslim annotators in the case of the upper plots).

tasks are subjective and typically aggregated based on majority-voting (Ivey et al., 2025; Giannoutsos et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025). This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems (Sap et al., 2019; van der Velden et al., 2025).

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These *fundamental* disagreements are typically ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). For example, coded language may not be picked up by most annotators but only by ones belonging in the targeted minority (Quaranto, 2022), as shown in Fig. 1. Furthermore, majority-group annotators may themselves be biased against minority groups, e.g., toxicity models disproportionately marking African American speech as “toxic” (Sap et al., 2019).

Instead of using inter-annotator agreement, we could use *polarization*, which formulates anno-

tation analysis as a cluster identification problem (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization (Pavlopoulos and Likas, 2024, 2023), there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question:

RQ. *Can we attribute polarization to annotator subgroups?*

We first identify limitations of current work on polarization:

1. Existing mechanisms (Pavlopoulos and Likas, 2024) operate only as binary classifiers, producing non-quantifiable/comparable results.
2. These mechanisms only test for the extreme case where *all* present polarization can be attributed to a single annotator group, which we find to be extremely rare in real-life datasets (App. B).
3. We are the first to report the existence of *inherent polarization*: most comments having some degree of polarization that cannot be attributed to *any possible partition* of annotators (Finding 1).
4. We also find that annotator groups may exhibit opposing polarization attribution between comments, requiring isolating annotations to each individual comment when analyzing whole datasets (Finding 2).

We introduce a new metric (“Aposteriori Unimodality–*apunim*”) that attributes polarization to specific annotator groups without and develop a parametric test for statistical significance. Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables *quantitative* comparisons across datasets. By considering the entire set of comments in a dataset *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets.

Having established our metric, we apply it to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection (§5), verifying current literature on the different annotation patterns of sociodemographic and ideological subgroups (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022). Finally, we release an open source Python library (Appendix D) that implements our metric, which can be directly installed via PyPi.¹ and provide an estimation on the minimum number of annotators required for robust estimates (§6). The code for the experiments, graph

¹Library: <https://pypi.org/project/apunim>
Documentation: <https://dimits-ts.github.io/apunim>

creation, and analysis can be found on our project’s repository.²

2 Related Work

2.1 Agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen’s kappa, Krippendorff’s alpha and Fleiss’s kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or “random disagreement”) using statistical methods. However, most cannot be generally used for comparing intergroup annotation agreement without extending them (e.g., computing pairwise Cohen’s kappa for all groups), or placing constraints on when they can be used (e.g., no missing labels and same number of annotations per annotator for Fleiss’s kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), for their inability to quantify and compare results across experiments (Wong et al., 2021), and for fundamentally faulty assumptions around the concept of “chance-correction” (Checco et al., 2017).

Recent work has attempted to work around these limitations, by working with simpler statistical models (van der Velden et al., 2025), adapting Cohen’s kappa for subgroup agreement (Wong et al., 2021), or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise (Ivey et al., 2025). While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive. They are also constrained by the same basic assumptions of disagreement metrics, such as not being suitable for multi-label annotations (where annotators can pick more than one option for each item).

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still necessitates traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms. This formulation, besides being appropriate for imbalanced groups, also enables metrics based on polarization to work with multi-label annotations, since we are no longer comparing indi-

²Replication code: <https://github.com/dimits-ts/aposteriori-unimodality>

vidual annotations against those of other annotators, but rather each annotator’s histograms (see §??). We use a non-parametric, normalized and intuitive approach (Pavlopoulos and Likas, 2024) as the basis for our metric.

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing polarization to subgroups is a relatively new idea. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are “causing” the exhibited polarization. Though promising, this idea is not practically usable, since subgroup polarization being exactly zero is an edge-case in real-life annotation tasks (Appendix B). Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.³

Such a binary analysis has been applied to discover whether Large Language Models (LLMs) are biased in the same way as specific human subgroups (Giannoutsos et al., 2025). This work shows that polarization is a versatile tool, but also highlights the constraints of using the non-quantifiable binary classification given by the traditional *aposteriori unimodality* framework. For instance, the authors have to resort to Cohen’s kappa agreement for any kind of comparison between LLMs and human annotator groups, constraining their analysis with the restrictions outlined above. The authors also show how ordinal dependent variables can extract “better” categorical labels via dataset-dependent thresholding, an idea that is promising yet limited (see §??).

3 Can we use polarization instead of agreement?

3.1 Problem Formulation

Annotations Let $d = \{c_1, c_2, \dots\}$ be a dataset composed of multiple annotated comments.⁴ Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \dots\}$, where a is a single annotation. As an example, the annotations for

³For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

⁴These items could be of any modality, such as comments in a discussion, images, videos or survey questions, even though we consider comments in this study.

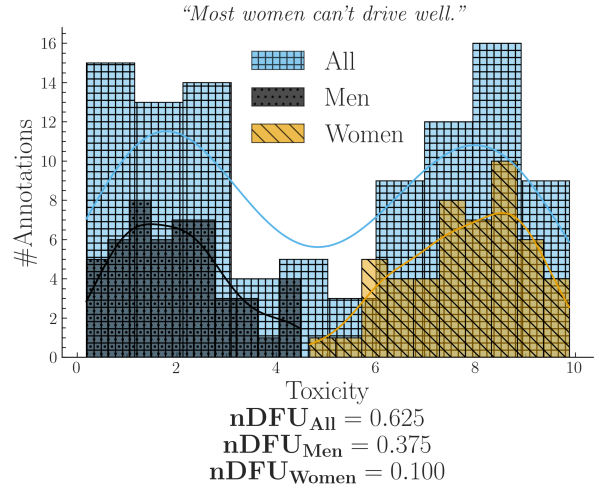


Figure 2: Synthetic experiment simulating a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{“Could be better, could be worse”}) \\ = \{ (+, F), (-, M), (+, F), \dots \}$$

Measuring polarization In the absence of polarization, annotation distributions are typically unimodal: annotators may disagree on finer details—shown as a bell-shaped distribution around a central mode—rather than on fundamentally different interpretations, which would instead give rise to multimodal distributions (Figure 2). We express unimodality through normalized Distance From Unimodality ($nDFU$) (Pavlopoulos and Likas, 2024), which takes values in the $[0, 1]$ range, with $nDFU \approx 0$ indicating no polarization (unimodal distributions) and $nDFU \approx 1$ corresponding to complete polarization (highly multimodal distributions). Our objective is therefore to quantify the extent to which the ($nDFU$) score of a polarizing item can be (partially) explained by a given Personal Characteristic (PC) group θ .

Polarization attribution Intuitively, a PC dimension partially explains the polarization of an item c when the annotations of that group $A(c|g) = \{a(c; g') \in A(c) | g' = g\}$ exhibit higher polarization compared to the full set of annotations $A(c)$. Figure 2 exhibits how polarization differs from agreement (left) and a hypothetical example where a misogynistic comment is annotated for toxicity by male and female annotators (right). The annotations are generally polarized ($nDFU_{all} = 0.62$). The annotations by female annotators exhibit low

polarization between themselves ($nDFU_{women} = 0.10$), since most agree that the comment is toxic. The set of male annotations, also shows low polarization ($nDFU_{men} = 0.37$), but for the opposite reason—most men agree that it is *not* toxic. This suggests that the overall polarization is driven by substantive disagreements between male and female annotators.

3.2 Inherent polarization

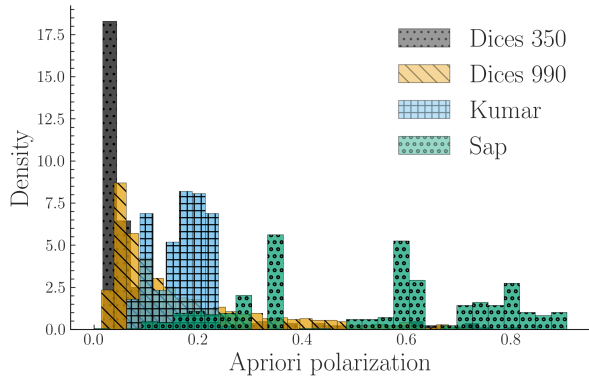


Figure 3: Apriori (inherent rather than observed) polarization across our datasets. Contrary to assumptions made in prior work, inherent polarization is rarely zero and can even reach $nDFU = 1$.

Inherent polarization refers to disagreement that cannot be explained by *any* grouping of annotators. It may arise from the task itself, the annotation instructions, ambiguity in the content, or simply random variation. Prior work (Pavlopoulos and Likas, 2024) attributed polarization to annotator groups only when the observed polarization for a grouping became zero, implicitly assuming that inherent polarization was also zero.

To examine this assumption, we create random annotator groupings for each comment in our datasets. Specifically, for every comment, we randomly partition the annotators into groups 100 times and compute the observed polarization for each partition.⁵ We then average these polarization scores across all random partitions. Repeating this process for every comment produces the distribution shown in Fig. 3. The results show that inherent polarization is consistently non-zero and can be substantial for many comments.

Finding 1. *Each comment may exhibit a degree of inherent polarization that cannot be attributed to any known or latent annotator dimension.*

3.3 Conflicting polarization

Previous work (Pavlopoulos and Likas, 2024) stopped on a comment-to-comment analysis. In practice however, this approach is inherently noisy due to a lack of information (i.e., limited annotations) relating to each individual comment. Many datasets in NLP are annotated by only 3–5 annotators per comment, which is especially problematic when we want to analyze patterns in annotator subgroups; in a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split. This issue exists across any subgroup analysis, including polarization. Since $nDFU$ is at its core a histogram-based estimation, three annotations may not be enough to acquire a reliable histogram, and thus estimate of polarization. In order to get meaningful information, we need to consider the annotations from all comments in a dataset.

However, *arbitrarily combining annotations would not work* in practice. Figure 4 illustrates a hypothetical discussion with two comments, both of which are toxic, but where the two groups of annotators disagree on *which* comment is the toxic one. If we aggregate the annotations to a single discussion, the opposing polarization effects might balance each other out, leading to a false negative. This demonstrates that item-level polarization attribution has a *direction*, which is not captured by $nDFU$, and has to be taken into account when aggregating annotations between items. This approach follows similar work which assumes different base annotation modes for each item (Checco et al., 2017).

Finding 2. *Polarization has a direction, which may compromise cross-comment annotation aggregation.*

4 Fixing polarization attribution

4.1 Group vs global polarization

In §??, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c|g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset. In this case, standard practice dictates that we should compare the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Figure 2, we would end up subtracting the $nDFUs$ of men and women, which are similar as they are both

⁵We assume two same-sized groups for this experiment. This assumption does not compromise the main finding of the experiment, but it is worth noting that we can use group sizes conditioned on the actual annotator group, as shown in §??.

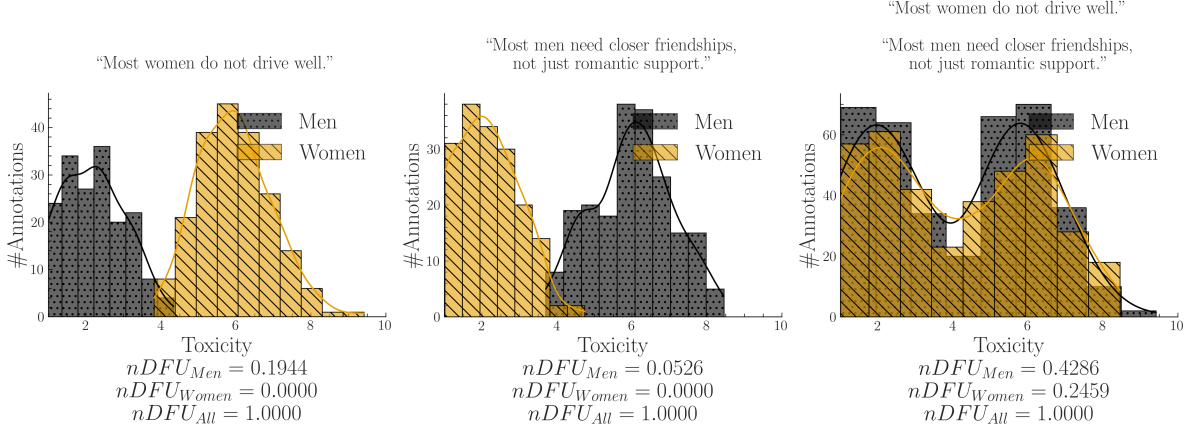


Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization.

unimodal distributions, concluding that there is no polarization present.

In order to bypass this restriction, we substitute calculating the observed polarization of $A(c)$ for the inherent polarization of the comment (§??). Specifically, we create random stratified splits that effectively “break up” the effect of any single isolated group on the polarization of the sample; i.e., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we create random partitions (disregarding gender) of sizes 80 and 20, which we repeat t times. Formally, the (average) apriori polarization for a single comment c is given by:

$$P_{apr}(c) = \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{r}_i(A(c)))$$

where $\tilde{r}$ is the random partition operator with group sizes matching the group sizes of the estimated groups.

4.2 Expanding to the whole dataset

Then, we can estimate how much polarization is generally present in the dataset, irrespective of any group, by simply averaging the *apriori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the dataset is likely not driven by any group.

$$P_{apr}^d = \frac{1}{|d|} \sum_{c \in d} P_{apr}(c) \quad (1)$$

Similarly to Equation 1, we also define polarization that was *observed* by an annotator subgroup as:

$$P_{obs}^d(g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c | g)) \quad (2)$$

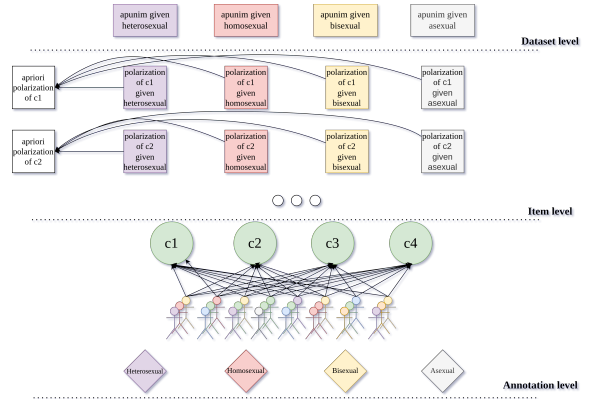


Figure 5: An overview of how annotation information is sequentially aggregated for the *apunim* metric. For each item we calculate the apriori polarization and the sample aposteriori polarization given each PC group. We then aggregate both apriori and aposteriori polarization and compare them on the dataset level to obtain the polarization attribution (“apunim”) values for each group.

This formulation implicitly assumes that dataset polarization can be estimated by averaging polarization scores across its items. We consider this assumption reasonable, because the mean is a simple and intuitive operator, exhibiting appropriate sensitivity to outliers. Given that polarization values lie in the $[0, 1]$ range, a single item with extreme polarization is unlikely to substantially affect the estimate of overall polarization. In contrast, the presence of many such extreme values will meaningfully increase this estimate, which is the pattern we want to capture.

We can now compare the polarization that is attributed to a single-dimensional annotator group

(Equation 2) to a prior one (Equation 1). We define our normalized metric, which we call as “apunim” (*A*posteriori *U*nimodality), as:

$$apunim(g) = \frac{P_{obs}^d(g) - P_{apr}^d}{1 - P_{apr}^d} \quad (3)$$

The scope of the statistic and its relationship with the previously presented statistics are demonstrated in Figure 5. Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- $apunim(\theta) \approx 0$: The exhibited polarization among annotators of group θ does not meaningfully differ from all other groups. In other words, it is indistinguishable from noise (represented by the apriori polarization).
- $apunim(\theta) > 0$: The polarization observed in the annotation task rises when this group is added to the annotation pool (equivalently, intragroup polarization is lower than apriori polarization). For example, if we observe $apunim(\text{“African American”}) = 0.3$, $apunim(\text{“White”}) = 0.001$ in a hate-speech detection task, it is likely that African American (AA) annotators fundamentally disagree with white annotators on what constitutes racism, regarding dataset d .
- $apunim(\theta) < 0$: The overall polarization drops when we add annotators from this group to this task. Mathematically, this occurs when intragroup polarization is higher than apriori polarization. This may indicate that annotators of that group may be very polarized between themselves. For example, if we observe $apunim(\text{Latino}) = -0.3$, $apunim(\text{White}) = 0.001$, it is likely that Latino annotators fundamentally disagree with each other, compared to AA or white annotators.

4.3 Statistical significance

Since the underlying annotation histograms may be noisy, we develop a statistical significance test. If the apunim value is statistically significant, we expect it to significantly differ from *apunim values exhibited by random annotator groups*. We use the individual randomized partitions $r_i(A(c))$ to compute pseudo-apunim values, which are then compared to the actual apunim value of the sample using a Student-T test. Thus, we test a much stricter hypothesis:

$$H_0 : apunim(g) = apunim(d)$$

$$H_a : apunim(d, g) \neq apunim(d)$$

where $apunim(d)$ denotes the mean apunim values of each random partition $\tilde{r}_i(A(c))$. For a detailed explanation of the p-value implementation, as well as a discussion around the assumptions made about the test, refer to App. C.

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. $\#Ann$ refers to the number of annotations per comment (see App. B), and $\#Dims$ to PC dimensions. Tasks are either toxicity (Tox) or hate speech detection (HS).

5 Attributing real-life polarization to human groups

5.1 Datasets

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar” (Kumar et al., 2021) and “Sap” (Sap et al., 2022) datasets, which are concerned with Toxicity and hate speech detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.⁶ While the Toxicity and hate speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

The datasets differ substantially in scale and annotation design. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. A brief overview of dataset characteristics is provided in Table 1. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (Fig. 8).⁷

5.2 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the toxicity / hate speech annotations, as shown in Table 1.⁸ Note that some groups are not included, either because

⁶For the latter, we consider only labels relating to racism (Q3_bias_overall), thereby transforming the task to hate speech detection.

⁷This systematic discrepancy could be attributed to the DICES datasets being composed of LLM-generated dialogues with already-aligned models tuned to avoid racist speech.

⁸We sample 20,000 comments from the Kumar dataset due to computational costs.

the results were not statistically significant, or because there were no polarized items annotated by more than one annotator of these groups (see §??). Consult Appendix E for details on other parameters and execution times, and Appendix F for the full results.

5.3 Which factors contribute to polarization?

Finding 3. *Polarization is consistently attributed to annotator race/ethnicity across all datasets, and especially so on hate-speech detection tasks*

Across all four datasets, we observe a consistent pattern in which annotation polarization can be attributed to differences in annotator ethnicity—in the hate speech detection datasets (DICES-350, DICES-990, Sap) this is the strongest contributor to polarization. In each case, statistically significant effects emerge for at least one of the major ethnic groups represented in the data (AAs: -0.0428 /Asians: 0.0659 for DICES-350, AAs: -0.2967 / Whites: 0.1400 for Sap, Whites: -0.0377 for Kumar, and all ethnicities for DICES-990). This verifies relevant literature (Goyal et al., 2022) suggesting that ethnic background constitutes a systematic source of annotator disagreement rather than isolated noise, and is more prevalent on tasks explicitly investigating racism.

Finding 4. *Some PC dimensions exert a non-zero-sum influence on the dataset.*

There is a general tendency for the *apunim* values of subgroups of the same PC to sum to zero. This indicates that overall, these PCs does not contribute to the polarization of this task, or equivalently, that its subgroups “cancel each other out”. Examples of this pattern can be found in Kumar for ‘‘Education’’, ‘‘Gender’’, ‘‘Political Affiliation’’, ‘‘Religion Important’’ and ‘‘Technology Impact’’. This does not mean that differences within these groups do not exist—indeed, our results verify relevant studies on the different annotation patterns for annotators of different gender (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020), and religious beliefs (Sachdeva et al., 2022).

Some PC dimensions however do not necessarily sum to zero, which suggests that *these dimensions may exert a systematic influence* on the total polarization of the dataset. In other words, some annotator characteristics can consistently increase or decrease observed polarization (§??). In the Kumar dataset, for example, respondents who report having been previously targeted contribute positively to polarization (0.0349), while those who have not been targeted contribute much more negatively (-0.0633). Similarly, transgender annotators contribute negatively to polarization (-0.0894) which is not balanced out by cisgen-

Group	apunim	support
DICES-350		
Race=AfricanAmerican	-0.043	9077
Race=Asian	0.066	8138
DICES-990		
Age=GenX+	-0.047	14384
Age=GenZ	0.064	13741
Edu=High school and below	-0.075	7419
Gender=Man	0.028	32494
Gender=Woman	-0.024	33471
Race=AfricanAmerican	0.091	5810
Race=Latino	-0.133	6610
Race=Other	0.075	24321
Race=White	-0.120	11392
Sap		
Eth=Black	-0.297	1041
Eth=White	0.140	2048
Gender=Male	0.094	1635
Gender=Female	-0.181	1333
Kumar		
Edu=Masters	0.015	13420
Edu=Bachelors	0.040	34085
Edu=Associate degree	-0.011	9122
Edu=None	-0.034	16666
Eth=White	-0.038	45306
Gender=F	-0.028	40686
Gender=M	0.028	37929
PastTargeted=No	-0.063	45234
PastTargeted=Yes	0.035	24491
Parent=No	-0.071	37231
Parent=Yes	0.054	41493
Trans=No	-0.089	11978
Politics=Conservative	0.033	23010
Politics=Independent	-0.016	22818
Politics=Liberal	-0.029	32917
ReligionImportant=No	-0.080	26008
ReligionImportant=Low	-0.015	10055
ReligionImportant=Mid	0.021	20169
ReligionImportant=High	0.049	27284
SeenToxicity=No	0.031	20185
SeenToxicity=Yes	-0.072	43265
SexOrientation=Bi	0.018	10005
SexOrientation=Het	-0.057	38048
TechImporant=Mid	-0.027	40477
TechImporant=High	0.021	22471
ToxProblem=Rare	0.018	14457
ToxProblem=Freq	-0.022	24774
ToxProblem=VeryFreq	-0.015	10656

Table 2: Statistically significant ($p < 0.05$) a posteriori unimodality results across all four datasets. High *apunim* values indicate that polarization can be attributed to this group, while negative values (gray rows) indicate that there is more polarization among annotators of that group compared to all annotators (see §4.2). Consult App. F for the full results.

der ones (0.0013). These asymmetries indicate that some characteristics shape the overall polarization of the dataset rather than simply redistributing it across groups.

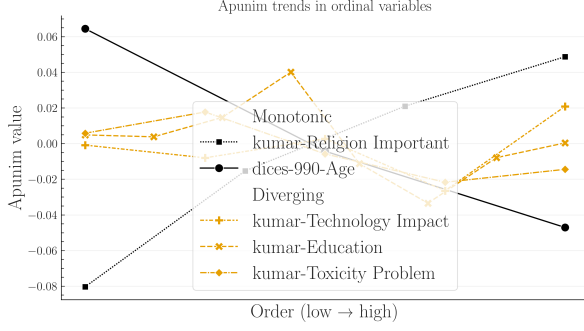


Figure 6: Apunim values across ordinal factor levels with at least two statistically significant values. The x-axis corresponds to the ordered levels of each factor and is normalized so that different ordinal scales are comparable (e.g., education which is measured in different scales across datasets). The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and indifference towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in F.

Finding 5. *Less educated annotators are more subjective.*

Education tends to have a moderate but consistent effect across the Kumar and DICES-990 datasets; the more educated an annotator is, the more they fundamentally disagree with other annotators. In the Kumar dataset, for instance, Bachelor’s-degree holders show a substantial positive contribution (0.0401), while groups with lower education such as “College, no degree” exhibit a negative contribution (−0.0335). The same pattern can be found in DICES-990, where annotators with at most a high-school education exhibit negative polarization attribution −0.0751, versus college-educated annotators whose values are extremely close to 0 (hence, why they are not shown in Table 2—the full results can be found in F). This may indicate that less educated annotators are more likely to exhibit subjective views, while educated annotators will tend to broadly agree with each other.

5.4 Why we should stop binarizing groups

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. One could treat all dimensions as *nominal/categorical*, even if they are inherently ordinal (e.g., LIKERT-scale, education level) or even binarize when many values exist (e.g., age). With this approach, however, not only do we lose valuable information, but we may magnify statistical errors (Streiner, 2002). By contrast, we posit that *retaining the order of ordinal attributes reveals in-*

teresting trends that would be otherwise missed with categorical inference.

Figure 6 shows *apunim* trends per ordinal PC attribute as shown in Table 2, focusing only on attributes with at least two statistically significant values ($p < 0.05$). This condition is necessary, since we cannot infer trends from dimensions with a single non-zero *apunim* value.⁹ We can see that **Religion Important** is characterized by a strict monotonic increasing trend. This indicates that non-religious annotators (low order) are more divided among themselves, while very religious annotators (high order) agree among themselves, but fundamentally disagree with other annotators. A very similar, inverse trend is observed for **Age**. For **Education**, we observe a peak in the middle (4—“Bachelor’s Degree”), which indicates two findings. First, up to an education level and as their level increases, people tend to agree more with people from a similar level. Second, beyond that level and till they reach the highest level, people tend to disagree more on their own level compared to ones from a different level.

6 How many annotators do we need?

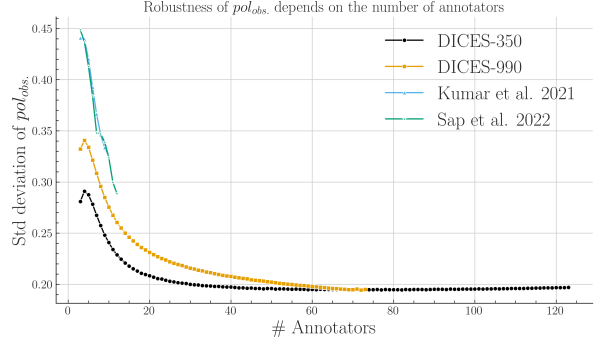


Figure 7: Standard deviation of 30 randomly sampled (with replacement) pol_{obs} values (Eq. 2). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (B).

While our test works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive (Rossi et al., 2024). Thus, it is imperative to find the least annotators needed for *apunim* to generate stable and reliable estimates. For each item per dataset, we randomly sample with replacement a subset of

⁹Note that **Age** is typically a ratio (positive, continuous) variable, although it is treated either as ratio (Sap), interval (Kumar), or ordinal (DICES) variable depending on the dataset. Since ratio and interval variables are also ordinals, we treat **Age** as ordinal in this analysis.

annotators and compute the standard deviation of 30 sampled pol_{obs} values (see Equation 2). Figure 7 demonstrates the effect of the number of annotators on the reliability of the polarization estimation, which we discuss next.

Each dataset features a different standard deviation even with the same number of annotators, which is broadly consistent with the amount of polarization in each dataset (Appendix B). However, we observe a clear, non-linear, inverse relationship between the number of annotators and the standard error across datasets. This can be attributed to a more robust histogram estimation used by nDFU (§??) when provided with more points for each of the subgroups. In the case of our datasets, we observe that after about 20 points, the estimation is good enough that more annotations contribute trivially to the robustness of the metric.

7 Conclusion

We introduced the *apunim* metric which attributes polarization to annotator subgroups for various PC dimensions. Our metric works well for imbalanced annotator splits, is well-defined for multi-label annotation, includes a statistical significance test, and unlike previous work, quantifies the amount of attribution in order to allow for comparisons between various experiments and datasets. Using this metric, we discover that annotator ethnicity is the strongest and most consistent driver of polarization between annotators across datasets, and is much more pronounced on hate speech detection tasks. We also discover that education and religion consistently impact the behavior of annotators. Specifically, we discover that very religious annotators fundamentally agree with themselves but disagree with others in Toxicity detection tasks, a trend that is gradually and consistently diminished, and ultimately reversed, by irreligious annotators. We also provide an open-source, PyPi-installable implementation, and an estimation of the number of annotators needed for each item, for our method to be consistently reliable.

8 Discussion

Future work Our framework has broad applications in NLP sub-tasks relating to content moderation (such as toxicity and hate speech detection). Since many such tasks are run on multiple discussions, our framework can be used on the discussion, instead of the dataset level. In that case, researchers may be able to isolate polarizing discussions, and investigate underlying patterns of polarization for individual minority groups. Alternatively, items can be grouped by comment turn, allowing researchers to investigate the evolution of polarization across discussions. Lastly, our work

can be used to more reliably test whether LLM-as-a-judge systems can replicate the polarization patterns of certain minority groups.

Limitations Researchers often only publish single-label, aggregated data, or datasets without annotator PC characteristics. Our analysis and the applicability of our framework is thus constrained by the lack of availability of such datasets. Additionally, as with related work, the substantial costs of human annotation prevent us from getting reliable estimates from an item-level analysis.

AI use Statement LLMs were used for minor styling changes in this document, for some implementation details in the Python framework, as well for some of the code used for figure generation. The authors have verified all LLM outputs.

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A Acronyms

LLM Large Language Model

PC Personal Characteristic

nDFU normalized Distance From Unimodality

FWER Family-Wise Error Rate

NLP Natural Language Processing

GPLv3 GNU General Public Licence v3

AA African American

B Annotation details

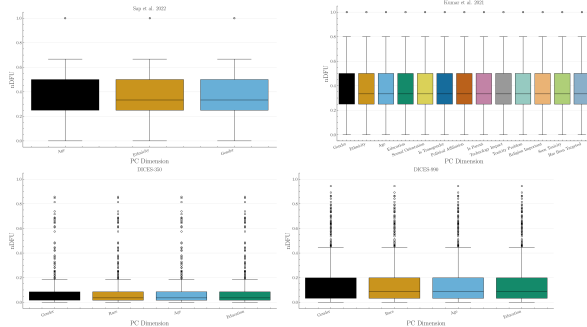


Figure 8: Histogram of dataset polarization for the human datasets considered in this paper per PC.

The number of annotators per comment for all datasets presented in §?? can be found in Figure 9, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 7 since they would produce uninterpretable figures.

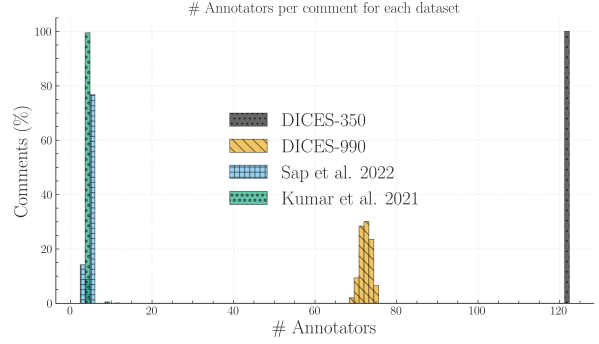


Figure 9: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3).

C P-value estimation implementation

Since our test is parametric, it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 1-2, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §??). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single θ dimension in principle, in practice we often want to test polarization for all subgroups ($\forall \theta \in \Theta$). Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%	75%	max
DICES-350	350	123	0	123	123	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73	74	76
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5	5	650
Sap et al. 2022	585	5.6359	0.7713	3	6	6	6	12

D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the `nDFU` and `apunim` calculations, as well as its statistical test. The software is available on PyPi (<https://pypi.org/project/apunim>). We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://dimitis-ts.github.io/apunim>).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project’s repository (<https://github.com/dimitis-ts/apunim>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy`, `scipy`, `statsmodels`).

D.2 Implementation details

By default, the computation of the `apunim` values happens on a single thread. In cases of serious computational cost, we recommend running the `apunim` calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple `PC` dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{P}_i$, which are originally calculated to obtain the apriori polarization values (see Equation 1), for the p-value calculations (§??). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{P}_i$ in Equation 1), and the number of `apunim` values used in the p-value calculation ($\tilde{D}_i$ in Algorithm ??) have to be the same.

E Replication Details

We use version 1.0.1 of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 1) and the p-value estimation (§??), and set a `FWER` rate of 0.95.

The current version of the library handles large number of annotators very well due to the use of vectorized operations, but struggles when faced

Group	apunim	support
Sap		
Age=GenX+	0.0192	743
Age=Millennial	0.0189	1792
Age=GenZ	−0.0153	239
Eth=Black	−0.2967	1041
Eth=White	0.1400	2048
Gender=Male	0.0942	1635
Gender=Female	−0.1810	1333

Table 4: Full `apunim` results for the Sap Dataset.

with a lot of comments when the annotators belong to multiple groups. Indeed, while the experiments for all datasets presented in §?? need no more than a minute to run, the Kumar dataset requires about 12 hours of computation even with a 20% sample. The annotator variance analysis presented in §6 also required more than 10 hours of computation, although that is expected due to the nature of the experiment, which is very computationally demanding, and a non-standard use of our library.

This bottleneck is likely caused by repeated creation of `numpy` arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each `PC` dimension. We plan on profiling and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each `PC` subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package (Tange, 2025) for our own experiments).

In Figure 7, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see B).

F Full Results

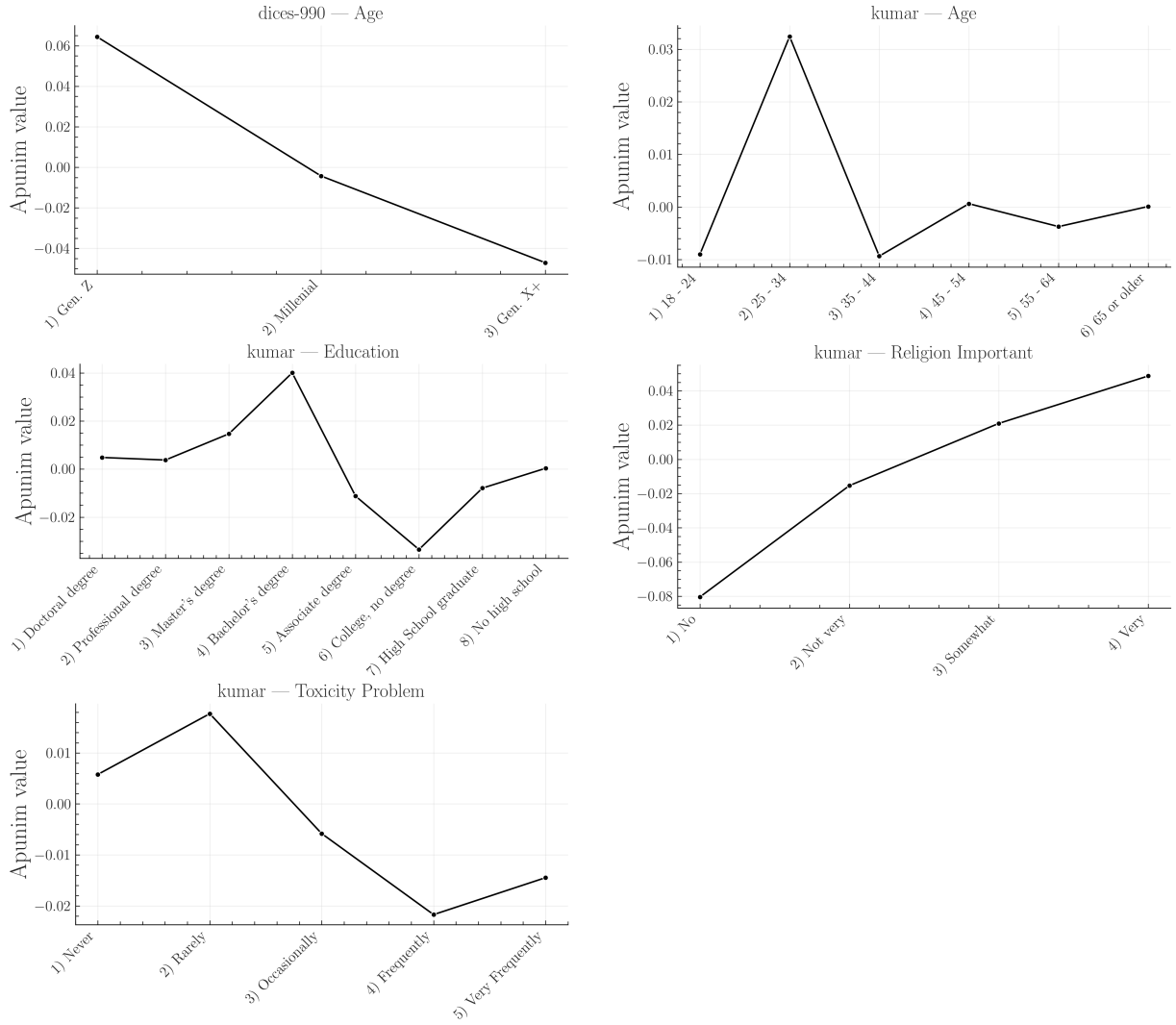


Figure 10: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal PC. This figure is complementary to Figure 6. The values in the graphs were extracted from the Tables of F.

Group	apunim	support
Kumar		
Age=1) 18-24	-0.009	9742
Age=2) 25-34	0.032***	33203
Age=3) 35-44	-0.009	21055
Age=4) 45-54	0.001	11025
Age=5) 55-64	-0.004	6123
Age=6) 65+	0.000	2567
Edu=1) Doctoral degree	0.005	1019
Edu=2) Prof. degree	0.004	1321
Edu=3) Master's degree	0.015**	13420
Edu=4) Bachelor's degree	0.040***	34085
Edu=5) Associate degree	-0.011**	9122
Edu=6) College, no degree	-0.034***	16666
Edu=7) High School	-0.008*	7326
Edu=8) No high school	0.000	462
Eth=Asian	-0.002	5062
Eth=Black	0.011*	10999
Eth=Hispanic	-0.003	2389
Eth=Multiracial	-0.004	3682
Eth=Other	0.002	1507
Eth=Unknown	-0.003	1160
Eth=White	-0.038***	45306
Gender=Female	-0.028***	40686
Gender=Male	0.028***	37929
Gender=Nonbinary	-0.001	489
Gender=Unknown	-0.003	621
Target=False	-0.063***	45234
Target=True	0.035***	24491
Parent=No	-0.071***	37231
Parent=Prefer not to say	-0.002	831
Parent=Yes	0.054***	41493
Trans=No	-0.089***	11978
Trans=Prefer not to say	0.000	856
Trans=Yes	0.001	2306
Pol=Conservative	0.033***	23010
Pol=Independent	-0.016***	22818
Pol=Liberal	-0.029***	32917
Pol=Other	-0.005	1822
Pol=Prefer not to say	0.002	3548
Rel=1) No	-0.080***	26008
Rel=2) Not very	-0.015***	10055
Rel=3) Somewhat	0.021***	20169
Rel=4) Very	0.049***	27284
Rel=Prefer not to say	0.000	1179

Table 5: Full apunim results for the Kumar Dataset.

Group	apunim	support
Kumar		
Tox=False	0.031***	20185
Tox=True	-0.072***	43265
SO=Bisexual	0.018***	10005
SO=Heterosexual	-0.057***	38048
SO=Homosexual	-0.002	2975
SO=Other	-0.000	728
SO=Prefer not to say	-0.002	1751
Tech=1) Very Low	-0.001	784
Tech=2) Low	-0.008	7973
Tech=3) Mid	0.003	10720
Tech=4) High	-0.027***	40477
Tech=5) Very high	0.021***	22471
ToxP=1) Never	0.006	4962
ToxP=2) Rarely	0.018***	14457
ToxP=3) Occasionally	-0.006	29476
ToxP=4) Frequently	-0.022***	24774
ToxP=5) Very Frequently	-0.015***	10656

Table 6: Full apunim results for the Kumar Dataset (cont'd).

Group	apunim	support
DICES-350		
Race=AfricanAm.	-0.0428	9077
Race=Asian	0.0659	8138
Age=Millenial	-0.0059	11268
Age=GenZ	-0.0066	17528
Edu=College or higher	0.0124	23475
Edu=High school or lower	-0.0152	12833
Edu=Other	-0.0214*	2191
Gender=Man	-0.0193	19093
Gender=Woman	0.0215	19406
Race=Latino	-0.0030	6886
Race=Multiracial	-0.0001	5008
Race=White	-0.0198	9390

Table 7: Full apunim results for the DICES-350 Dataset.

Group	apunim	support
DICES-990		
Age=GenX+	-0.0471	14384
Age=Millenial	-0.0043	38436
Age=GenZ	0.0644	13741
Edu=College or higher	0.0095	59142
Edu=High school or lower	-0.0751	7419
Gender=Man	0.0278	32494
Gender=Woman	-0.0235	33471
Race=AfricanAm.	0.0909	5810
Race=Latino	-0.0030	6886
Race=Multiracial	-0.0001	5008
Race=White	-0.0198	9390

Table 8: Full apunim results for the DICES-990 Dataset.